HOLY CROSS COLLEGE

Life to the Full

12 PHYSICS

THE STANDARD MODEL

The Standard Model is based on the theories of **quantum mechanics** and **special relativity** and the assumption that interactions between particles are caused by exchange particles called **gauge bosons**.

Three of the four fundamental forces of nature are described by the quantum field theories of ***quantum electrodynamics*** and ***electroweak theory***.

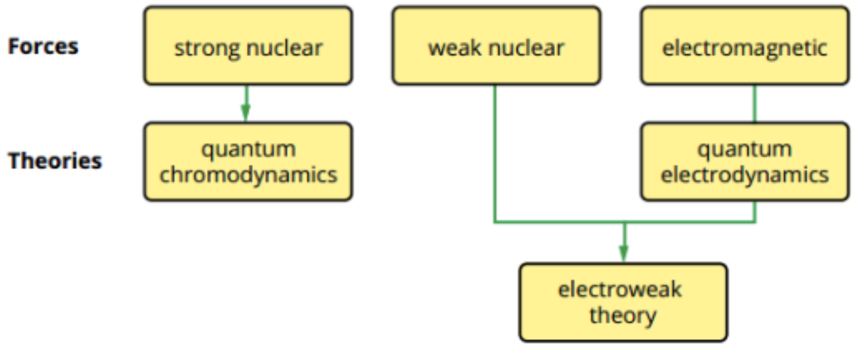


FIGURE 9.1.2 The three fundamental forces that are described by the quantum field theories of **quantum chromodynamics** and the electroweak theory.

electrons
neutrinos

Physicists have been able to classify the particles they have discovered into three groups - **leptons**, **gauge bosons** and **hadrons**, according to their interactions.

The diagram below illustrates how the known particles are classified under the Standard Model.

interact or mediate forces

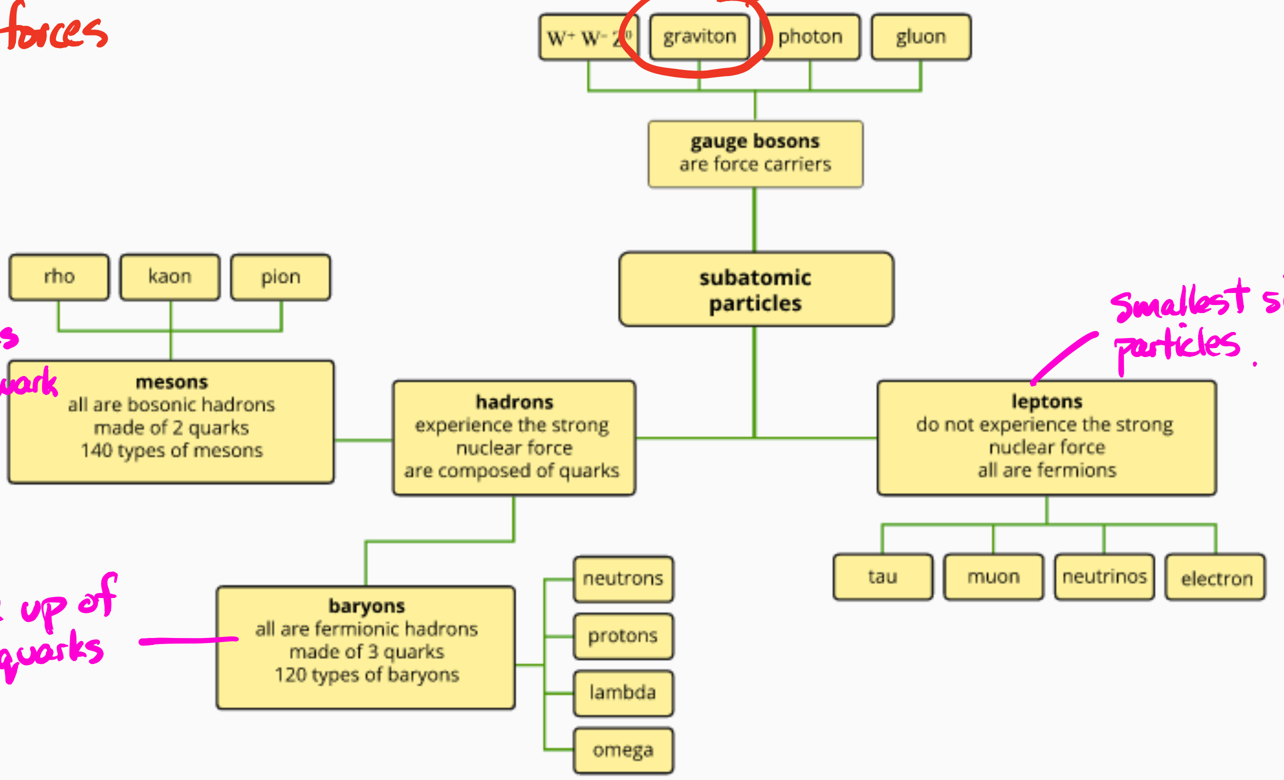
protons, neutrons, etc

still to be found

Made up of 2 quarks
- quark-antiquark pair

smallest single particles.

made up of 3 quarks



9.1.3 • This mind map summarises the classification of particles in the Standard Model of particle physics.

Fundamental Forces and The Gauge Bosons

There are four forces that can act on particles and govern their behaviour. The fundamental assumption in the Standard Model is that ***forces arise through the exchange of particles called gauge bosons (or just bosons).***

Bosons are often called force-carrying, force-mediating or exchange particles.

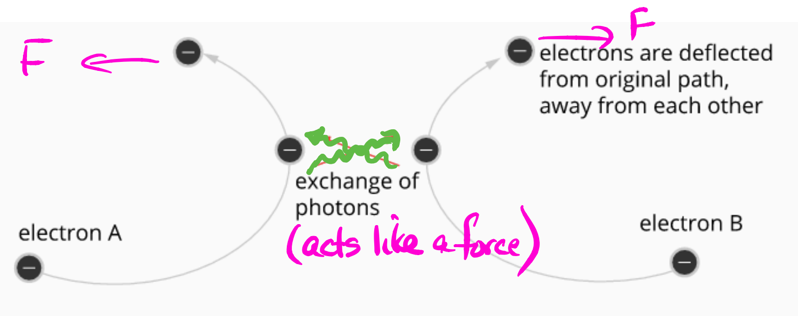
The table below gives a summary of the nature of these particles, their strength and the range over which they can exert a force.

A question has been asked in WACE exam on this table.

Force	Nature	Relative strength	Range (m)	Force carrier (gauge bosons)
strong nuclear	holds the nucleus of atoms together and acts between quarks	1	10^{-15} (~ diameter of an atomic nucleus)	gluon
electromagnetic	responsible for both electric and magnetic fields exerting forces of attraction or repulsion	$\frac{1}{137}$	infinite	photon
weak nuclear	causes radioactive decay	10^{-6}	10^{-18} (much less than the diameter of a proton)	W^+, W^- and Z
gravity	a force of attraction between any two objects with mass	6×10^{-39}	infinite	graviton (theoretical and unobserved)

Example

The diagram below represents an electron approaching another electron. Each electron emits a photon that is absorbed by the other. This causes each electron to experience a force of repulsion, and the photon is responsible for the electromagnetic force acting on the two electrons.



9.1.5 • Two electrons approach each other, are repelled and then move away from each other. The two electrons exchange a photon, which is the carrier of the electromagnetic force.

EXTENSION

Why gravity is not part of the Standard Model

The Standard Model of particle physics successfully describes the strong nuclear, electromagnetic and weak nuclear forces. Unfortunately, there is no well-developed and experimentally proven quantum theory of gravity. If this existed it might be able to be combined with the electroweak theory and quantum chromodynamics (the theory that describes the interactions of the strong force) to create a grand unified theory (GUT) that could explain the fundamental behaviour of all matter in the universe.

The behaviour of objects due to the force of gravity is described with spectacular accuracy by Einstein's general theory of relativity. This theory basically says that mass tells space how to bend, and bent space tells mass how to move. It applies on the largest scales in the universe and even predicts how binary black holes should emit gravitational waves.

Quantum mechanics describes mathematically the behaviour of subatomic particles. This theory covers the interactions of matter and energy that occur on the smallest scales and is one of the foundations of the Standard Model of particle physics.

Einstein's theory of general relativity does not work for small-scale dimensions and high energies. Subatomic particles exist on very small scales and in many cases, are at very high energies. Therefore, the Standard Model of particle physics cannot accommodate gravity and its theoretical exchange particle, the graviton.

MATTER

Matter is not as simple as protons, neutrons and electrons. These are not the smallest particles - Physicists have discovered a large number of smaller particles that make up matter.

For the purpose of this course, only a simplified view will be used.

Matter And Anti-Matter

For each particle that exists, there is an anti-matter particle to match.

e.g. An electron has its anti-matter particle - the positron. The positron has many of the same properties including mass, but it has a positive charge as opposed to the negative charge of the electron. Other features of the positron are also opposite to the electron's properties.

Once the positron was discovered, other anti-matter particles began to be uncovered, such as the anti-proton and the anti-neutron.

Anti-matter is extremely rare on Earth and is greatly outnumbered in the universe by regular matter, which is a puzzle.

When normal matter interacts with its anti-matter equivalent, the two particles will convert all of their mass into energy ($E = mc^2$) in a process called ***annihilation***.

Example

Calculate the energy released when an electron meets a positron and they are annihilated.

Solution

An electron has a mass of 9.11×10^{-31} kg; the anti-matter positron has many properties that are opposite to the electron, but it has the same mass, therefore the energy created when the two meet is:

$m_{e^-} = 9.11 \times 10^{-31}$ kg

$m_{e^+} = 9.11 \times 10^{-31}$ kg

$$E = mc^2$$
$$= (9.11 \times 10^{-31} + 9.11 \times 10^{-31})(3.00 \times 10^8)^2$$
$$= 1.64 \times 10^{-13}\ \text{J}$$

This energy is usually in the form of an emitted photon with a wavelength of 1.21×10^{-12} m, which places it in the gamma ray section of the electromagnetic spectrum. This is also the minimum wavelength photon that could create an electron–positron pair; a higher energy photon would result in the pair having some kinetic energy.

There are two conventions used to indicate that a particle is an antiparticle.

- The antiparticles of uncharged particles are indicated by placing a bar above the symbol for the normal matter particle. For example, an electron neutrino has the symbol ν_e and the antielectron neutrino has the symbol $\bar{\nu}_e$.
- The antiparticle of a charged particle is given the symbol of the particle but with the opposite sign. For example, the antiparticle of the muon (μ^-) is the antimuon (μ^+).

In the diagram right, an antiproton travelling along path L collided with a proton and they annihilated. This released a huge amount of energy and produced several new particles whose tracks are shown in the image.

FIGURE 9.2.3 This image records a proton–antiproton annihilation. This event was recorded in 1955 in a photographic emulsion at the Bevatron accelerator at the Lawrence Berkeley Laboratory, California.

Pair-production is the opposite of annihilation and is observed when energy in the form of a photon is used to ceate a particle-antiparticle pair. This can only occur if the eneergy of the photon is equal or greater than the combined mass of the pair of particles.

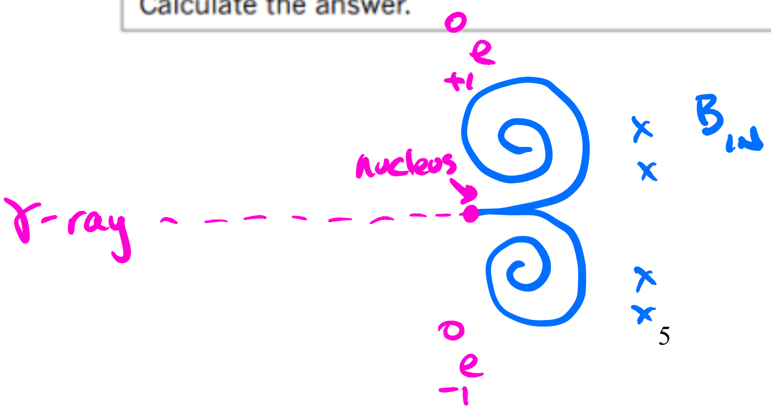
Example

$E = mc^2$

$\Rightarrow m = \frac{E}{c^2} = 0.511\ \text{MeV}/c^2$

CONSERVATION OF MASS–ENERGY

A 2 MeV gamma-ray photon interacts with an atomic nucleus and an electron–positron pair is produced. Each particle created has a mass of 0.511 MeV. How much kinetic energy is carried away by the particle–antiparticle pair produced?	
Thinking	**Working**
The energy of the gamma-ray photon is conserved and is equal to the total mass–energy of the particles produced plus the kinetic energy (E_k) of those particles.	$2\ \text{MeV} = 2 \times 0.511\ \text{MeV} + E_k$
Solve for the kinetic energy.	$2\ \text{MeV} = 1.022\ \text{MeV} + E_k$
Calculate the answer.	$E_k = 0.978\ \text{MeV}$



Pair production inside a cloud chamber

Classifying New Particles Read this but know the summary.

- ***Bosons***

  This group of particles ***mediate the strong nuclear, electromagnetic and weak forces***.

  They include the ***gluons, photons and the W and Z particles***.

- ***Leptons***

  This group of ***particles interact by exchanging W and Z bosons, which mediate the weak nuclear force***.

  Leptons that carry a charge can also interact by exchanging photons, which mediate the electromagnetic force, but they do not interact via the strong force carriers.

  These particles include the ***electron, muon and tau particles***, as well as their corresponding ***neutrinos*** and the ***anti-matter opposites of these six particles***.

- ***Hadrons***

  This group of particles is further categorised into two groups is, the ***baryons*** and the ***mesons***, based on a property called ***baryon number (B)***

  The common feature of the hadrons is that they ***all interact by exchanging gluons, which are the particles that mediate the strong nuclear force***.

  Hadrons that carry a charge can also interact by exchanging photons, but the effect of the gluons far outweighs any other force mediating particle.

- ***Mesons***

  These are particles that have been ***assigned a baryon number of zero***.

  In this group are many particles and their anti-particles.

  e.g. pion (π^+), anti-pion (π^-) and pi-zero (π^0), the kaon (K^+) and anti-kaon (K^-), and the eta (η^0).

- ***Baryons***

  These are particles that have been ***assigned a baryon number of 1 for normal matter or -1 for anti-matter***.

  In this group are the familiar ***proton*** (p^+) and ***anti-proton*** (p^-), ***neutron*** (n) and ***anti-neutron*** ($\bar{n}$), along with hundreds of other particles and their anti-particles.

  e.g. the lambda-zero (Λ^0) and anti-lambda-zero ($\bar{\Lambda}$), sigma-plus (Σ^+), sigmazero (Σ^0) a and sigma-minus (Σ^-), the xi-zero (Ξ^0) and omega-minus (Ω^-).

As far as physicists know, ***all six leptons are considered to be fundamental particles as there appears to be no internal structure to them*** and their size appears to be unmeasurable.

Hadrons on the other hand appear to have an internal structure, indicating that they are actually made of smaller particles. ***These smaller particles must then be considered as the fundamental particles in nature.***

These smaller particles are called ***quarks***.

The six different types of quarks are referred to as the six ***flavours*** of quarks - up (u), down (d), strange (s), charmed (c), bottom (b) and top (t).

The last four quark names also apply to new quantum numbers and their conservation laws, called strangeness (S), charm (c), bottomness (b) and topness (t).

Quarks have their antimatter opposites called ***anti-quarks*** that have the ***opposite sign*** for all of their quantum numbers.

Quarks have properties of ***baryon number***, ***spin*** and ***charge***, which must add together to give the total baryon number, spin and charge of the hadrons they combine to make.

Quarks also have another property called ***colour charge***, which is similar to electric charge, and distinguishes between two quarks of the same flavour, but with different spin. The three colours of quarks are red, green and blue and the colours for the anti-quarks are anti-red (cyan), anti-green (magenta) and anti-blue (yellow).

Quarks must combine so that their ***colour charge property adds to equal white.***

e.g. red + green + blue = white
red + cyan = white
green + magenta = white
blue + yellow = white

Name	Symbol	Charge (Q)	Baryon number (B)	Strangeness (S)	Charm (c)	Bottomness (b)	Topness (t)
Up	u	$+\frac{2}{3}$ e	$\frac{1}{3}$	0	0	0	0
Down	d	$-\frac{1}{3}$ e	$\frac{1}{3}$	0	0	0	0
Strange	s	$-\frac{1}{3}$ e	$\frac{1}{3}$	−1	0	0	0
Charmed	c	$+\frac{2}{3}$ e	$\frac{1}{3}$	0	+1	0	0
Bottom	b	$-\frac{1}{3}$ e	$\frac{1}{3}$	0	0	−1	0
Top	t	$+\frac{2}{3}$ e	$\frac{1}{3}$	0	0	0	+1

proton $^{1}_{1}p = u + u + d$

$= +\frac{2}{3} + \frac{2}{3} - \frac{1}{3}$

$= +1$

neutron $^{1}_{0}n = u + d + d$

$= +\frac{2}{3} - \frac{1}{3} - \frac{1}{3}$

$= 0$

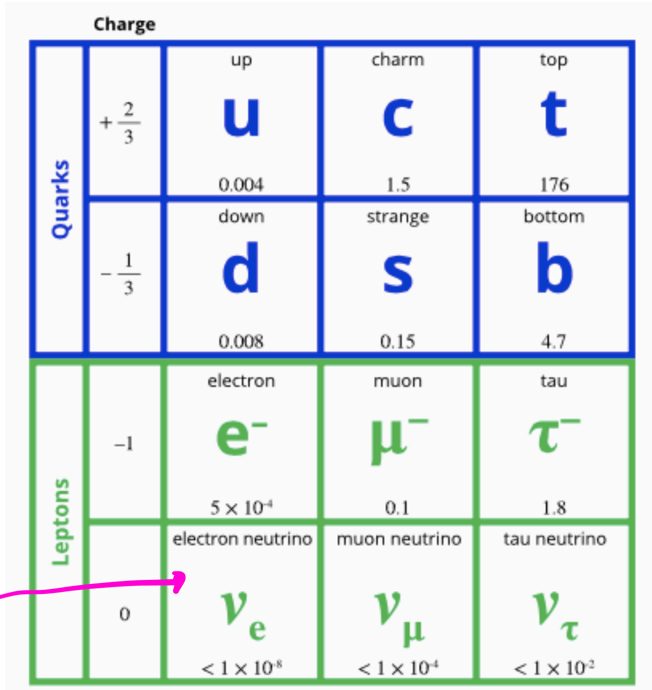
Fundamental Particles

The Standard Model says that all particles of matter are comprised of one or more of the 12 fundamental or elementary particles.

Being a fundamental particle means that it is not comprised of other smaller particles.

All these particles are called ***fermions*** and are divided into two groups of six particles called ***quarks*** (blue) and ***leptons*** (green).

electron has its associated neutrino



Note: The number below the symbol for each particle is the mass and is given relative to the mass of a proton, ~1 GeV/c^2.

All fermions (quarks and leptons) obey Pauli's exclusion principle. This states that no two particles can occupy the same quantum state at the same time.

> e.g. No two electrons can be in the same shell or orbital around an atom and have the same energy.

Fermions also have the same property, called ***spin angular momentum***, abbreviated to just 'spin'.

All fundamental fermions have a spin of +1/2.

All quarks experience the strong nuclear force and this separates them from the leptons, as they do not.

Quarks also have non-integer (fractional) charges, unlike leptons, which have charges of -1, 1 or 0.

All fermions that carry a charge can interact via the electromagnetic force. Since neutrinos have zero charge, they obviously do not experience this force.

Physics file: The Higgs boson

Until the 1960s, physicists still had no answer to the question, 'How do particles get their mass?'. British physicist Peter Higgs, and others, proposed an answer to this question in 1964. Using the Standard Model of particle physics, they predicted that there was a yet undiscovered particle that would later be known as the Higgs boson. This particle would interact with other particles and give them what we measure as mass.

Finally, in 2008, over 10 000 physicists and engineers collaborated to build the Large Hadron Collider (LHC) near Geneva, Switzerland. This is the largest and most complex experimental facility ever built and one of its primary goals was to test the prediction of the existence of the Higgs boson.

The search began and it was not until 2012 that the discovery of a candidate for the Higgs particle was announced. On December 10, 2013, two of the original researchers, Peter Higgs and François Englert, were awarded the Nobel Prize in Physics for their work and prediction.

Written as $m = \frac{E}{c^2}$

STANDARD MODEL OF ELEMENTARY PARTICLES

How To Make A Baryon

Baryons, which include the proton and neutron, ***consist of three quarks***.

e.g. ***Proton*** - made up of two up quarks and a down quark (uud).

Total charge = +1 (the charge on a proton).

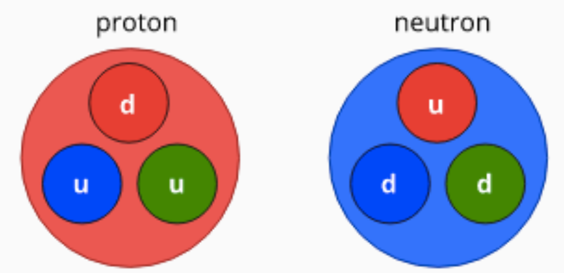
$$\text{proton} = \text{uud} = +\tfrac{2}{3} + \tfrac{2}{3} - \tfrac{1}{3} = +1$$

Neutron - made up of two down quarks and an up quark (ddu).

Total charge = 0

$$\text{neutron} = \text{ddu} = -\tfrac{1}{3} - \tfrac{1}{3} + \tfrac{2}{3} = 0$$

Charge adds up to 0.



9.1.8 • Protons and neutrons are baryons, which are all made of three quarks.

Anti-particles have the opposite Baryon number

Baryon number

The following table summarises the baryons and their quarks.

Name	Symbol	B	S	c	b	t	Quarks
Proton	p	+1	0	0	0	0	uud
Anti-proton	$\bar{p}$	−1	0	0	0	0	$\overline{uud}$
Neutron	n	+1	0	0	0	0	udd
Anti-neutron	$\bar{n}$	−1	0	0	0	0	$\overline{udd}$
Lambda-plus	Λ^+	+1	0	+1	0	0	udc
Lambda-zero	Λ^0	+1	−1	0	0	0	uds
Sigma-plus	Σ^+	+1	−1	0	0	0	uus
Sigma-zero	Σ^0	+1	−1	0	0	0	uds
Sigma-minus	Σ^-	+1	−1	0	0	0	dds
Xi-zero	Ξ^0	+1	−2	0	0	0	uss
Xi-plus	Ξ^+	+1	−2	0	0	0	dss
Omega-minus	Ω^-	+1	−3	0	0	0	sss

How To Make A Meson

Mesons consist of a ***quark and an anti-quark pair***.

e.g. The pion-plus (π^+) consists of an up quark and an anti-down quark ($u\bar{d}$).

Total charge = +1

$$\text{pion - plus} = u\bar{d} = +\tfrac{2}{3} + \tfrac{1}{3} = +1$$

Name	Symbol	*B*	*S*	*c*	*b*	*t*	Quarks
Pion-plus	π^+	0	0	0	0	0	$u\bar{d}$
Pion-minus	π^-	0	0	0	0	0	$\bar{u}d$
Kaon-plus	K^+	0	+1	0	0	0	$u\bar{s}$
Kaon-minus	K^-	0	−1	0	0	0	$\bar{u}s$
Rho-plus	ρ^+	0	+1	0	0	0	$u\bar{d}$
Rho-minus	ρ^-	0	−1	0	0	0	$\bar{u}d$
phi	φ	0	0	0	0	0	$s\bar{s}$
D-plus	D^+	0	0	+1	0	0	$c\bar{d}$
D-zero	D^0	0	0	+1	0	0	$c\bar{u}$
D-plus-s	$D^+{}_s$	0	+1	+1	0	0	$c\bar{s}$
B-minus	B^-	0	0	0	−1	0	$b\bar{u}$
Upsilon	Υ	0	0	0	0	0	$b\bar{b}$

Strong Nuclear Force And The Colour Charge

According to the theory of quantum chromodynamics (QCD), the force that binds the quarks together in a hadron is called the ***colour force*** and it acts between the colour charge of the quarks.

The strong nuclear force between nucleons (or between any hadrons) is therefore a result of the colour force between the quarks within each of the hadrons.

e.g. The attraction between one proton and another, within a nucleus, is a result of the ***attraction between the quarks*** contained within each proton.

This leads to the ***gluon particles that are exchanged in the strong nuclear force being the same particles that exchange the colour force***.

Particle Characteristics

The following tables show the various properties of the common leptons, mesons and baryons.

Note

Photons are not listed in the table - they are ***field particles*** and are involved in interactions between particles.

Lepton numbers

Table 9.1 Some particles and their properties

Category	Particle name	Symbol	Anti-particle	Mass (MeV/c^2)	B	L_e	L_μ	L_τ	Lifetime (s)	Spin
Leptons	Electron	e^-	e^+	0.511	0	+1	0	0	Stable	$\frac{1}{2}$
	Electron-neutrino	ν_e	$\bar{\nu}_e$	<7 eV/c^2	0	+1	0	0	Stable	$\frac{1}{2}$
	Muon	μ^-	μ^+	105.7	0	0	+1	0	2.20×10^{-6}	$\frac{1}{2}$
	Muon-neutrino	ν_μ	$\bar{\nu}_\mu$	<0.3	0	0	+1	0	Stable	$\frac{1}{2}$
	Tau	τ^-	τ^+	1 784	0	0	0	+1	$<4 \times 10^{-13}$	$\frac{1}{2}$
	Tau-neutrino	ν_τ	$\bar{\nu}_\tau$	<30	0	0	0	+1	Stable	$\frac{1}{2}$
Hadrons										
Mesons	Pion	π^+	π^-	139.6	0	0	0	0	2.60×10^{-8}	0
		π^0	Self	135.0	0	0	0	0	0.83×10^{-16}	0
	Kaon	K^+	K^-	493.7	0	0	0	0	1.24×10^{-8}	0
		K_S^0	$\bar{K}_S^0$	497.7	0	0	0	0	0.89×10^{-10}	0
		K_L^0	$\bar{K}_L^0$	497.7	0	0		0	5.2×10^{-8}	0
	Eta	η	Self	548.8	0	0	0	0	$<10^{-8}$	0
		η'	Self	958	0	0	0	0	2.2×10^{-10}	0

Category	Particle name	Symbol	Anti-particle	Mass (MeV/c^2)	B	L_e	L_μ	L_τ	Lifetime (s)	Spin
Baryons	Proton	p	$\bar{p}$	938.3	+1	0	0	0	Stable	$\frac{1}{2}$
	Neutron	n	$\bar{n}$	939.6	+1	0	0	0	614	$\frac{1}{2}$
	Lambda	Λ^0	$\bar{\Lambda}^0$	1115.6	+1	0	0	0	2.6×10^{-10}	$\frac{1}{2}$
	Sigma	Σ^+	$\bar{\Sigma}^-$	1189.4	+1	0	0	0	0.80×10^{-10}	$\frac{1}{2}$
		Σ^0	$\bar{\Sigma}^0$	1192.5	+1	0	0	0	6×10^{-20}	$\frac{1}{2}$
		Σ^-	$\bar{\Sigma}^+$	1197.3	+1	0	0	0	1.5×10^{-10}	$\frac{1}{2}$
	Delta	Δ^{++}	$\bar{\Delta}^{--}$	1230	+1	0	0	0	6×10^{-24}	$\frac{3}{2}$
		Δ^+	$\bar{\Delta}^-$	1231	+1	0	0	0	6×10^{-24}	$\frac{3}{2}$
		Δ^0	$\bar{\Delta}^0$	1232	+1	0	0	0	63×10^{-24}	$\frac{3}{2}$
		Δ^-	$\bar{\Delta}^+$	1234	+1	0	0	0	6×10^{-24}	$\frac{3}{2}$
	Xi	Ξ^0	$\bar{\Xi}^0$	1315	+1	0	0	0	2.9×10^{-10}	$\frac{1}{2}$
		Ξ^-	$\bar{\Xi}^+$	1321	+1	0	0	0	1.64×10^{-10}	$\frac{1}{2}$
	Omega	Ω^-	Ω^+	1672	+1	0	0	0	0.82×10^{-10}	$\frac{3}{2}$

Summary

Know this!

Hadrons

- Made up of 2 or three quarks.
- Interact via the strong nuclear, electromagnetic, gravitational and weak forces.
- Divided into mesons and baryons

2 quarks

Mesons

- Have integer spins (0, 1, 2, ...).
- Are bosons.
- No stable meson exists - they all decay.
- Meson decay produces electrons, positrons, neutrinos and photons.

Baryons

- Mass equal or greater than a proton.
- Have half-integer spins (1/2, 3/2).
- Protons and neutrons are baryons - protons are the only stable ones.

Leptons

- Don't interact via the strong nuclear force - only the other three.
- All have spin 1/2.
- Have no internal structure and are point-like.

Field particles

- The four fundamental forces are mediated by a field particle or exchange particle.

Table 9.2 Forces and their exchange particles

Force	Exchange particle
Electromagnetic	Photon, γ
Gravitational	Graviton (not yet detected)
Weak	W and Z bosons, W^+, W^-, Z^0
Strong nuclear (not a fundamental force)	Pions, π^+, π^-, π^0

Conservation of Charge and Quantum Numbers

The quantum number for ***charge*** is conserved in particle decays.

Example

A physicist proposed that she had observed a collision of a proton and a neutron, and the products were two protons and an antiproton. Is this reaction allowed according to the law of conservation of charge? 2 parts

Thinking	Working
To find out if this reaction is allowed, calculate the total charge present before and after the interaction.	$p^{+} + n \rightarrow p^{+} + p^{+} + p^{-}$ Proton's charge $= +1$ Neutron's charge $= 0$ Antiproton's charge $= -1$ Total charge before $= +1 + 0 = +1$ Total charge after $= (+1) + (+1) + (-1) = +1$
Compare the total charge before and after the interaction. If they are equal then the reaction is allowed; if they are different then the reaction is forbidden.	The total charge before and after is the same. Therefore, this proposed reaction is allowed according to the law of conservation of charge.

	charge
LHS	$+1 + 0 = +1$
RHS	$+1 + 1 - 1 = +1$

$\therefore$ charge conserved

Conservation of Baryon Number

Baryons are all composite particles containing combinations of three quarks.

e.g. Protons and neutrons.

All baryons are given a quantum number - ***B = +1.***

For antibaryons - ***B = −1.***

For all of other particles involved in the interaction or decay - ***B = 0.***

Example

▲ A physicist proposed that she had observed a collision of a proton and a neutron and the products were two protons and an antiproton. Is this reaction allowed according to the law of conservation of baryon number? 2 parts

Thinking

The law of conservation of baryon number states that the net number of baryons remains constant in any process. To find out if this reaction is allowed, calculate the total baryon number before and after the interaction.

Working

$$p^+ + n \rightarrow p^+ + p^+ + p^-$$

Proton $B = +1$

Neutron $B = +1$

Antiproton $B = -1$

	Baryon
LHS	+1 +1 = +2
RHS	+1 +1 −1 = +1

Total B before $= (+1) + (+1)$
$= +2$

Total B after $= (+1) + (+1) + (-1)$
$= +1$

∴ Not possible

Thinking

Compare the total baryon number before and after the interaction. If they are equal then the reaction is allowed according to the law of conservation of baryon number. If they are different then the reaction is forbidden as baryon number is not conserved.

Working

The total baryon number before the reaction is different to the total baryon number after the reaction. Therefore, this proposed reaction is forbidden according to the law of conservation of baryon number.

Conservation of Lepton Number

Leptons are the fundamental particles shown below.

Charge	Leptons		
–1	electron e^-	muon μ^-	tau τ^-
0	electron neutrino ν_e	muon neutrino ν_μ	tau neutrino ν_τ

Lepton number, L, is a quantum number with value of ***+1 for leptons***, ***−1 for antileptons*** and ***0 for other particles***.

There are actually ***three types of lepton numbers***: one type for the electron and the electron neutrino, $\boldsymbol{L_e}$; one type for the muon and the muon neutrino, $\boldsymbol{L_\mu}$; and one type for the tau and the tau neutrino, $\boldsymbol{L_\tau}$.

Each of these three lepton numbers must be considered and conserved independently. Both must be conserved for the reaction to be allowed.

$L_{electrons}$ is conserved
L_{muons} " "
L_{tau} " "
Treat them individually.

Example

A physicist proposed that she had observed the decay of a muon (μ^-) into an electron (e^-), an antielectron neutrino ($\overline{\nu_e}$) and a muon neutrino (ν_μ). Is this reaction allowed according to the law of conservation of lepton number? 5 parts

$$\mu^- \rightarrow e^- + \overline{\upsilon}_e + \upsilon_\mu$$

Not part of $L_{electron}$

	$Lepton_\mu$	$Lepton_e$
LHS	$+1$	0
RHS	$0+0+1=+1$	$+1-1+0=0$

$\therefore$ Possible

Thinking	Working
Identify which types of leptons are involved in the interaction. These need to be considered separately.	electron/electron neutrino, L_e: yes muon/muon neutrino, L_μ: yes tau/tau neutrino, L_τ: no
To find out if this reaction is allowed, calculate the total electron lepton number L_e before and after the interaction.	Electron $L_e = +1$ Antielectron neutrino $L_e = -1$ Total before $L_e = 0$ Total after $L_e = (+1) + (-1) = 0$
Compare the total electron lepton number before and after the interaction. If they are equal then the reaction is allowed. If they are different then the reaction is forbidden.	The total electron lepton number before the reaction is the same as the total electron lepton number after the reaction. Therefore, this proposed reaction is allowed according to the law of conservation of electron lepton number.

Thinking	Working
To find out if this reaction is allowed, calculate the total muon lepton number L_μ before and after the interaction.	Muon $L_\mu = +1$ Muon neutrino $L_\mu = +1$ Total before $L_\mu = +1$ Total after $L_\mu = +1$
Compare the total muon lepton number before and after the interaction. If they are equal then the reaction is allowed. If they are different then the reaction is forbidden.	The total muon lepton number before the reaction is the same as the total muon lepton number after the reaction. Therefore, this proposed reaction is allowed according to the law of conservation of muon lepton number.

INTERACTIONS BETWEEN PARTICLES

Interactions between particles that produce some change and are observed by a particle detector are commonly known as ***events***.

Those interactions that result in the formation of new particles are sometimes referred to as ***reactions***.

Interactions can involve attractive or repulsive forces, decay or annihilation.

Examples

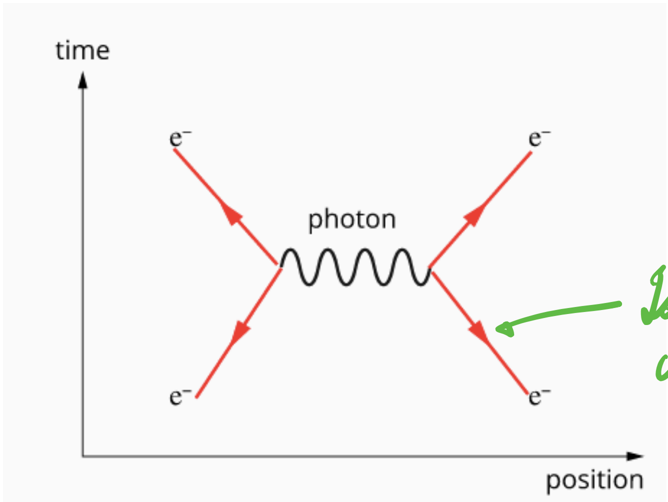
An example of a ***repulsive interaction*** involving the electromagnetic force would be the scattering of one electron by another.

This example is illustrated below using a diagram called a ***Feynman diagram***.

Not part of the course???

It shows two electrons approach each other (on the horizontal axis) over time (the vertical axis) and exchange a photon.

This causes each electron to be repelled and therefore change direction.



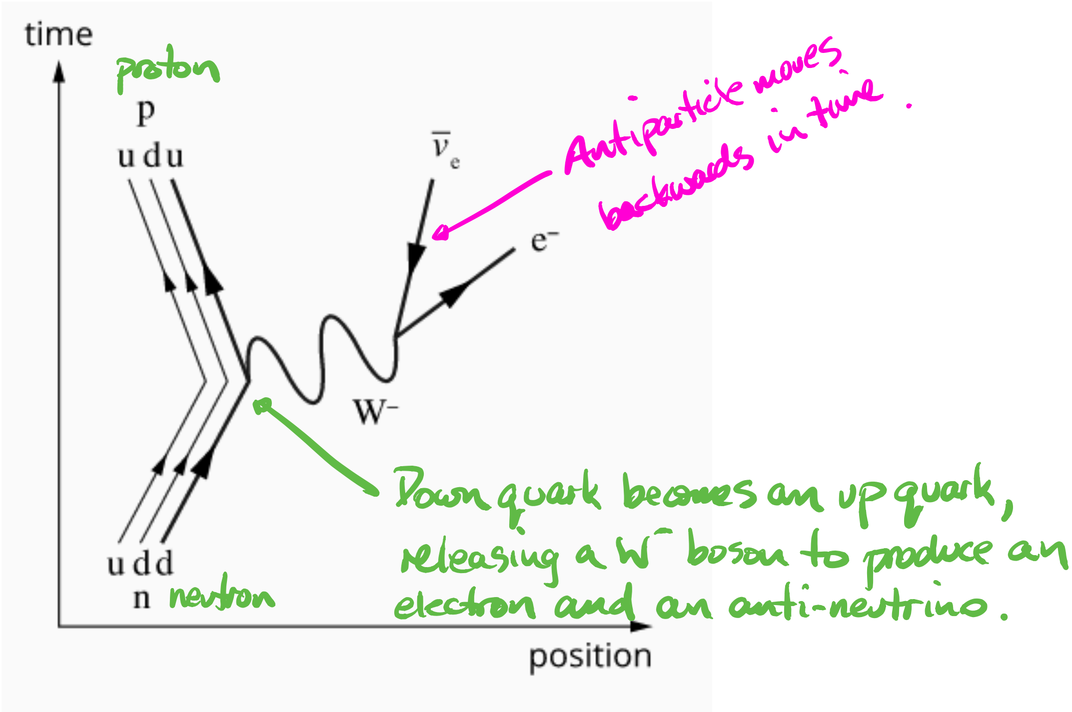
Is this arrow in the correct direction?

An example of a ***decay*** is the decay of a neutron into a proton.

This involves the weak nuclear force and is brought about by the conversion of a down quark in the neutron into an up quark.

This requires the emission of a W− boson, which decays into an antielectron neutrino and an electron.

This is commonly known as a ***beta-minus radioactive decay***.



Note

It is important to note that the directions of the arrows in these diagrams do not indicate the direction a particle travels.

e.g. ***An antiparticle moves backwards in time.***

Particles are represented by straight lines - the direction of the arrow represents the direction of time.

For antiparticles - the direction is reversed.

Wiggly lines represent massless exchange particles such as photons.

ASTRONOMY

Distances in Space

The only way the astronomers who followed Galileo could measure the distance to the stars was to look for parallax in their positions as the Earth moved around the Sun.

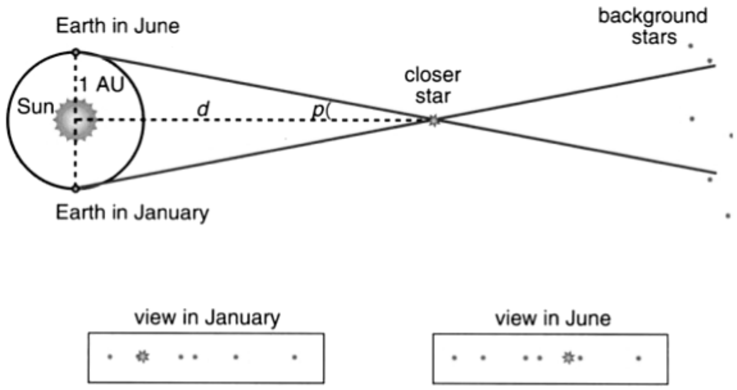


Figure 6.36
The parallax movement of a close star relative to the background 'fixed' stars. The parallax angle p is half the total apparent shift in angle.

As the Earth moves around its orbit the closer star should appear to move in relation to those further away.

This method of measuring the distance to stars is known as ***stellar parallax***. It leads to a natural unit for the distance of stars, the ***parsec***.

The parsec (abbreviation pc) is the distance to a star that shows a parallax angle of 1.0 arc second.

Physics file

What is a parsec (pc)? Astronomers define the **parallax angle** as the angle subtended by the radius of the Earth's orbit (1 AU) as it would be seen at the distance of the star—as shown in Figure 6.37. The total circumference of a circle of radius r would be $2\pi\rho$.

As 1 arcsec is $\frac{1}{3600}$ of a degree, and a degree $\frac{1}{360}$ of a full circle, we can see that 1 AU must be:

$$\frac{1}{3600 \times 360} = \frac{1}{1\,296\,000}$$

of the full circumference; that is:

$$1 \text{ AU} = \frac{2\pi r}{360 \times 3600}$$

You can show that this leads to $r = 206\,265$ AU, which is indeed the conversion factor from pc to AU.

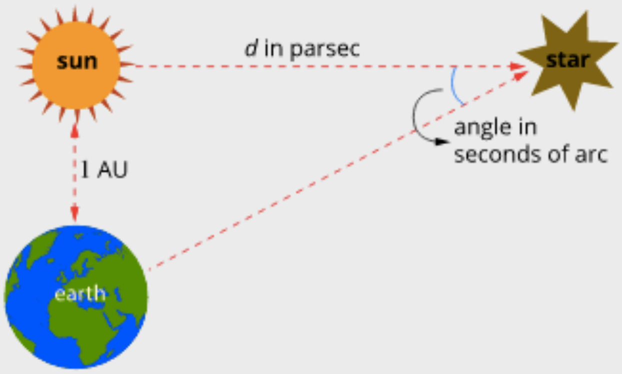
Physics file: Units of distance in space

It is hard to picture the vast range of distances in space. Astronomers use two measurements in particular.

The 'astronomical unit', or AU, is the distance from the Earth to the Sun. It is equivalent to 1.5×10^8 km. It is useful when discussing the distances within the solar system or to nearer stars. For example, the distance to Pluto is about 40 AU, while that to the nearest star (Proxima Centauri) is about 270 000 AU.

For larger distances the parsec is used. This is the distance you would have to be from the Earth so that the radius of the Earth's orbit around the Sun would make an angle of one second of arc. It is equivalent to 206 265 AU. The kiloparsec (kpc, 10^3 pc) and megaparsec (Mpc, 10^6 pc) are also used for galactic distances.

Modern telescopes have been used to measure the distances of galaxies containing Cepheids out to about 30 Mpc (megaparsec). Using newer techniques, astronomers have now found galaxies out to around a thousand megaparsecs.



9.4.4 • This diagram illustrates the relationship between astronomical units (AU) and parsecs (pc).

There is a simple relationship between a star's distance and its parallax angle:

$$\boxed{d = \frac{1}{p}}$$

where d = distance in parsecs
p = parallax angle in arcseconds.

Note

1.00 parsec = 3.26 light years

On data sheet.

Limitations of Distance Measurement Using Stellar Parallax

Parallax angles of less than 0.01 arcsec are very difficult to measure from Earth because of the effects of the Earth's atmosphere. This limits Earth-based telescopes to measuring the distances to stars about 1/0.01 or 100 parsecs away.

Space-based telescopes can get accuracy to 0.001, which has increased the number of stars whose distance could be measured with this method. However, most stars even in our own galaxy are much further away than 1000 parsecs, since the Milky Way is about 30,000 parsecs across.

Example

A star has a parallax angle p of 0.723 arseconds. What is the distance to the star in parsecs and light years?

$$\begin{aligned} d &= \frac{1}{p} \\ &= \frac{1}{0.723} \\ &= 1.38 \text{ parsecs} \\ &= \underline{4.50 \text{ light years}} \end{aligned}$$

More Distant Stars

Cepheid variable stars (very bright stars that vary in brightness over a period from 1-2 days to several weeks) and supernovae can be used to measure larger distances, such as the distances between galaxies. By comparing the intrinsic brightness to the apparent brightness, Edwin Hubble was able to measure the distances to stars and galaxies far away.

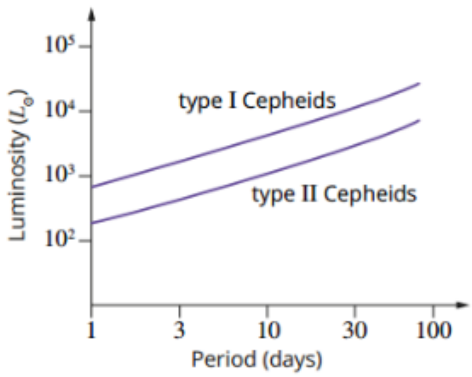


FIGURE 9.4.2 The period–luminosity relationship for Cepheid variables. There are two types that can be distinguished by the metal content of their spectra. $L_\odot$ is the luminosity of the Sun. Fortunately, Cepheids are very much brighter than the Sun, enabling them to be seen even in distant galaxies.

MODELS OF THE UNIVERSE

Aristotle (ancient Greece) - Earth in centre of universe.

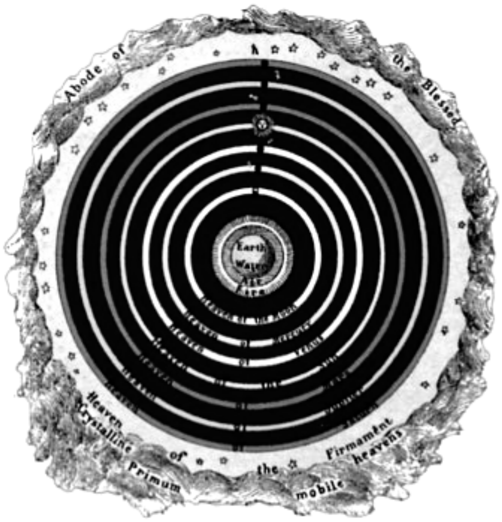


Figure 6.53

Aristotle taught that the Earth was at the centre of the Universe. It was surrounded by water, air and fire. Beyond that was the heavenly realm of the Moon, the Sun and the planets, which described perfect circles around the Earth. The celestial sphere, which surrounded all of this, was the region of the stars and the home of the gods.

Ptolemy (200 BC) - Earth in centre; planets moved in circles around a large circle.

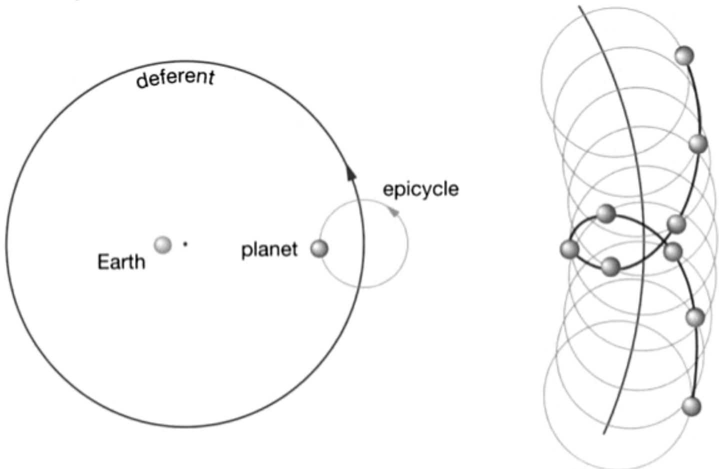


Figure 6.56

Ptolemy's system of planetary motion based on epicycles, the centre of which moved on a circle called the deferent centred near, but not on, the Earth. When on the inside of the deferent, the planet could exhibit retrograde motion as shown.

Copernicus (16th century) - Sun in centre, Earth and other planets circle it.

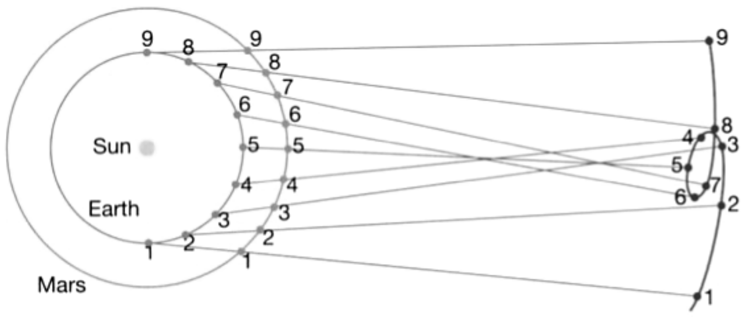


Figure 6.57
Using the Copernican explanation of retrograde motion, Mars appears to move backwards for a couple of months each year when the Earth 'overtakes' the more slowly moving planet.

Kepler (1600) - formulated that planets orbit in ellipses with the Sun at one focus. He developed three laws to describe planetary motion.

Kepler's First Law states that the orbit of a planet is an ellipse with the Sun at one focus.

Kepler's Second Law states that the radius from the Sun to a planet sweeps out equal areas in equal times.

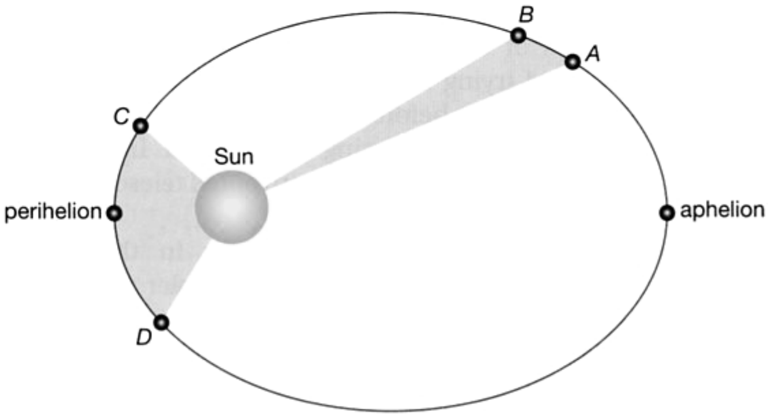


Figure 6.64
Kepler's second law says that a planet sweeps out equal areas in equal times. This means that it is moving faster at perihelion (closest to the Sun) and slowest at aphelion (furthest from the Sun). This ellipse is exaggerated; the orbits of most planets are quite close to circular.

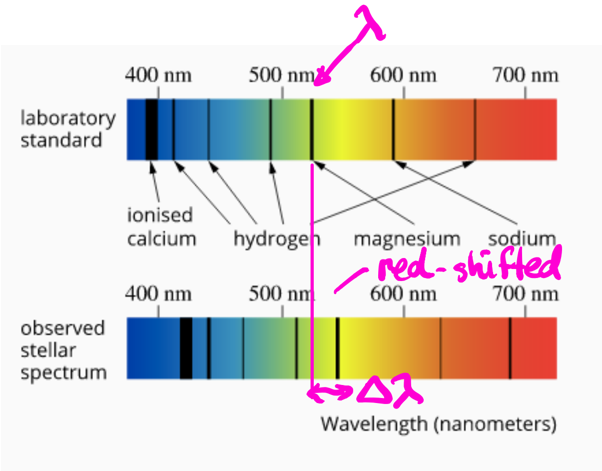
Kepler's Third Law states that the square of the period of the planet (T^2) is proportional to the cube of the radius of the orbit (R^3). That is:

$$\boxed{\frac{R^3}{T^2} = K}$$

where *K* is Kepler's constant.

HUBBLE'S UNIVERSE

In the 1920's, Edwin Hubble was making detailed observations of galaxies. He found that when he looked at the spectrum of a galaxy, it was ***"red shifted".*** This means that it had the same pattern of lines as normal spectra but they were all moved towards the red (long wavelength) end of the spectrum.



9.4.5 • The spectrum of a star in our galaxy is redshifted, indicating it is moving away from the Earth in its motion around the galaxy.

$z = \frac{\Delta\lambda}{\lambda}$

This was interpreted as the well-known 'Doppler shift' that occurs whenever a source of waves is moving towards or away from us.

The speed of the galaxy can be found from the amount of red shift.

Not on the data sheet.

$$\boxed{v_{galaxy} = \frac{\Delta\lambda}{\lambda}c}$$

$z = \frac{v_{galaxy}}{c} = \frac{\Delta\lambda}{\lambda}$

where z = amount of red shift

where v_{galaxy} = the speed of the observed galaxy (ms^{-1}).
$\Delta\lambda$ = the change in wavelength (m).
λ = the normal wavelength (m).
c = the speed of light (ms^{-1}).

Hubble measured the distance of the galaxies and found a remarkable pattern. The more distant galaxies had greater red shifts, and the red shift was directly proportional to the distance of the galaxy.

i.e. The galaxies seemed to be speeding away from us, and those at the greater distances were receding at greater speeds.

This became known as ***Hubble's Law*** and is expressed as a simple equation.

$$\boxed{v_{galaxy} = H_o d}$$

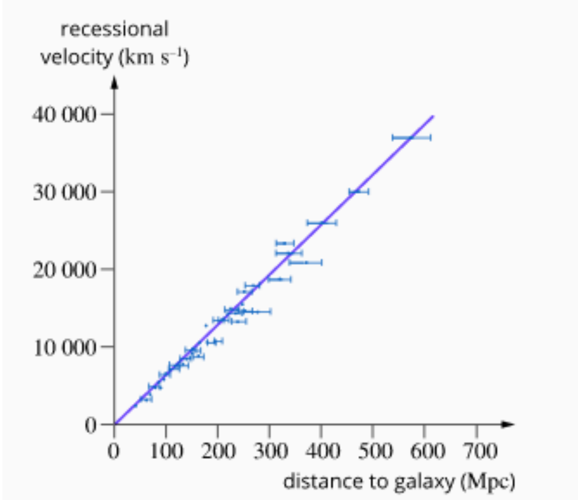
Not on the data sheet

where v_{galaxy} = the recession velocity of the galaxy (kms^{-1}).
H_o = Hubble's constant ($kms^{-1}\ Mpc^{-1}$).
d = the distance of the galaxy (Mpc).

The value of Hubble's constant is about 70 kms^{-1} Mpc^{-1}.

It has proved difficult to find an accurate value for the constant because of the huge distances of the galaxies involved, but recent results from a satellite called WMAP (Wilkinson Microwave Anisotropy Probe) suggest a value of 71 kms^{-1} Mpc^{-1} to within 5%.

Not all galaxies are receding from us. Some of the closer ones are moving towards us (and so have ***blue shifts***) owing to their gravitational interactions.



9.4.6 • Hubble discovered that the galaxies were receding from us at velocities proportional to their distance. The slope of the line is known as Hubble's constant, H_0.

Example

The spectral line of singly ionised calcium has a wavelength of 393.3 nm when measured in the laboratory, but can be seen to have a wavelength of 401.8 nm in the light from the elliptical galaxy NGC 4889.

a Find the recession speed of this galaxy.

b Use Hubble's law to find the distance to this galaxy.

$$z = \frac{\Delta\lambda}{\lambda} = \frac{(401.8-393.3)\times10^{-9}}{393.3\times10^{-9}}$$

$$= \underline{\qquad\qquad}.$$

Solution

a

$\lambda_{lab} = 393.3\times10^{-9}$ m $\qquad v_{galaxy} = \frac{\Delta\lambda}{\lambda}c$

$\lambda_{obs} = 401.8\times10^{-9}$ m $\qquad = \frac{(401.8\times10^{-9})-(393.3\times10^{-9})}{393.3\times10^{-9}}(3.00\times10^{8})$

$c = 3.00\times10^{8}$ m s^{-1} $\qquad = 6.48\times10^{6}$ m s^{-1}

$\qquad = 6.48\times10^{3}$ km s^{-1}

$$z = \frac{\Delta\lambda}{\lambda} = \frac{v_{galaxy}}{c}$$

$$\Rightarrow v_{galaxy} = \frac{\Delta\lambda c}{\lambda}$$

$$= \ldots$$

$$= \underline{\qquad\qquad}.$$

b $H_o = 70$ km s^{-1} Mpc^{-1} $\qquad v_{galaxy} = H_o d$

$v_{galaxy} = 6.48\times10^{3}$ km s^{-1} $\qquad d = \frac{v_{galaxy}}{H_o} = \frac{6.48\times10^{3}}{70}$

$= 92.6$ Mpc

$= (92.6\times10^{3})(3.262\times10^{3})$

$= 3.02\times10^{8}$ l.y.

This distance is 302 million light-years and so we are seeing the galaxy as it was 302 million years ago.

Physics in action — The Doppler effect

When a source of waves, whether sound, light or any other, is moving towards an observer, the apparent wavelength seen by the observer will be shorter (girl in Figure 6.71a). This is because each wave is emitted a little closer to the observer than the previous one and so is not so far behind the previous wave as it would be if the source was stationary. The reverse is the case if the source is moving away from the observer (boy in Figure 6.71a).

In Figure 6.71b two consecutive waves from a source moving with speed u are shown. The second wave has just been emitted at the position shown. The natural wavelength is λ and the period T. As the velocity of the wave is v, the wavelength $\lambda = vT$. In the time between emitting the two waves the source has moved a distance uT. This means that the distance between the waves in the forward direction is $vT - uT$, while that in the backward direction is $vT + uT$. These distances are the wavelengths as perceived by the observers to the front and rear, respectively.

In the case of light waves travelling from moving sources, we are interested in the apparent change in wavelength $\Delta\lambda$. As can be seen in Figure 6.71b, this is simply uT, so $\Delta\lambda = uT$. As T is also given by λ/v, we can rewrite this expression as:

$$\Delta\lambda = \frac{u}{v}\lambda$$

That is, the wavelength change, or 'shift' in the terminology of astrophysics (redshift or blueshift depending on the direction), is equal to the wavelength times the ratio of the speed of the source to the speed of the wave. If we replace v by the speed of light waves, c, and u by the speed of the galaxy, v_{galaxy}, and then do a little algebraic reorganisation the expression can be written as:

$$v_{galaxy} = \frac{\Delta\lambda}{\lambda}c$$

So the speed of the galaxy is equal to the fractional wavelength shift ($\frac{\Delta\lambda}{\lambda}$) times the speed of light.

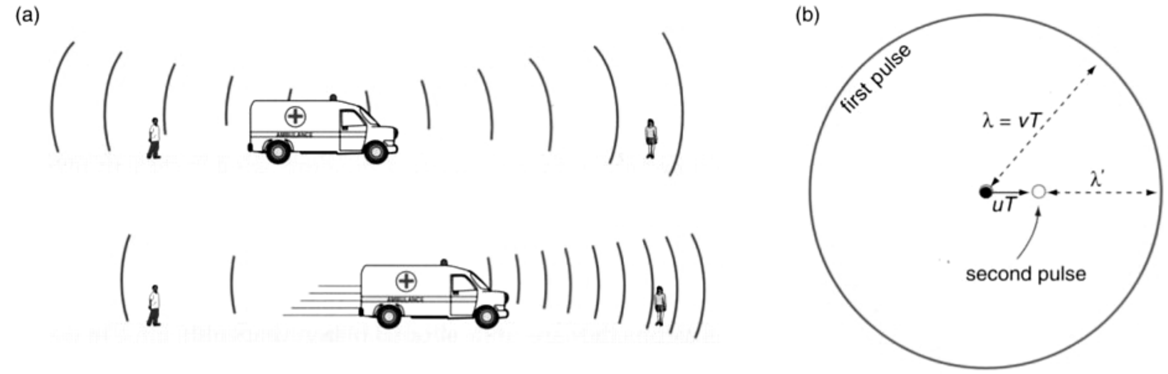


Figure 6.71
The Doppler effect.

EXTENSION

Cosmological redshift

It is worth noting here that although the redshift seen in distant galaxies was originally thought to be a Doppler shift—the result of motion away from us—it is more correctly interpreted as the direct consequence of expanding space, as shown in Figure 9.4.7. The wavelength of the radiation increased with space rather than in space. This type of redshift is more correctly called a 'cosmological redshift' and was predicted by Einstein's general theory of relativity.

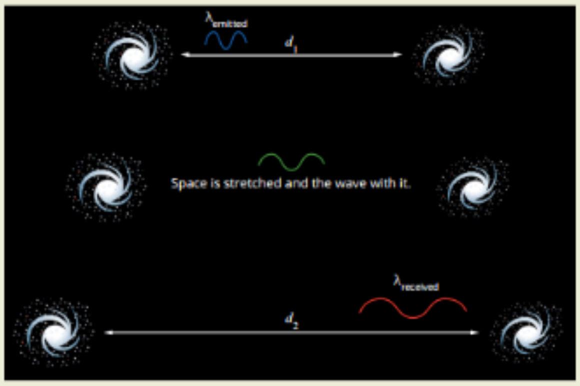


FIGURE 9.4.7 Light from distant galaxies is cosmologically redshifted during its journey through expanding space. The farther the light has travelled, the greater the redshift.

ORIGIN OF THE UNIVERSE

The Universe was thought to be infinite and unchanging in the time of Galileo and Newton.

Hubble realised that, because all the distant galaxies were receding from us, unless we just happened to be in the 'real' centre of the Universe, the whole universe must be expanding and therefore the light from extremely distant stars will never reach us.

It is not the galaxies all moving away from one point in space - it is ***space itself that is expanding***.

Analogy

Imagine an ant on a balloon that is being blown up. This balloon happens to have had little paper stars stuck onto it (not drawn on it). The ant would see all the paper stars moving away from it. Wherever the wanders on the balloon's surface, it would see exactly the same thing.

The ant can travel in a "straight" line and end up back where it started if we held rigidly to what we believed was a straight course through space (and given more time than the lifetime of the Universe!) eventually end up back where we started.

i.e. The universe would appear "closed".

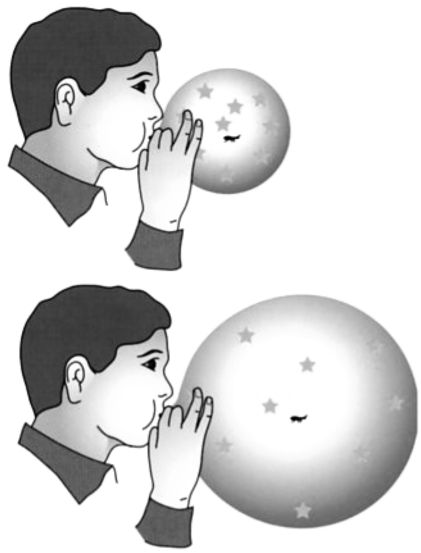


Figure 6.75
As the balloon is blown up, the stars all move away from the ant. The more distant stars will move away faster than the closer ones. In the real universe it is the galaxies that are moving away from each other.

It is important to make the distinction between this expansion of space and a conventional explosion, in which everything flies out from one point in space.

To see uniform expansion in the explosion one would have to be right at the centre.

This is not the case for the expansion of the Universe (or the ant on the balloon). Astrophysicists talk of this as the ***cosmological principle***. This states that the Universe is ***isotropic and homogeneous*** - it will appear uniform in whatever direction one look and from wherever.

Did it all start with a big bang?

If space is expanding, then at some time in the past it must have been just a dot.

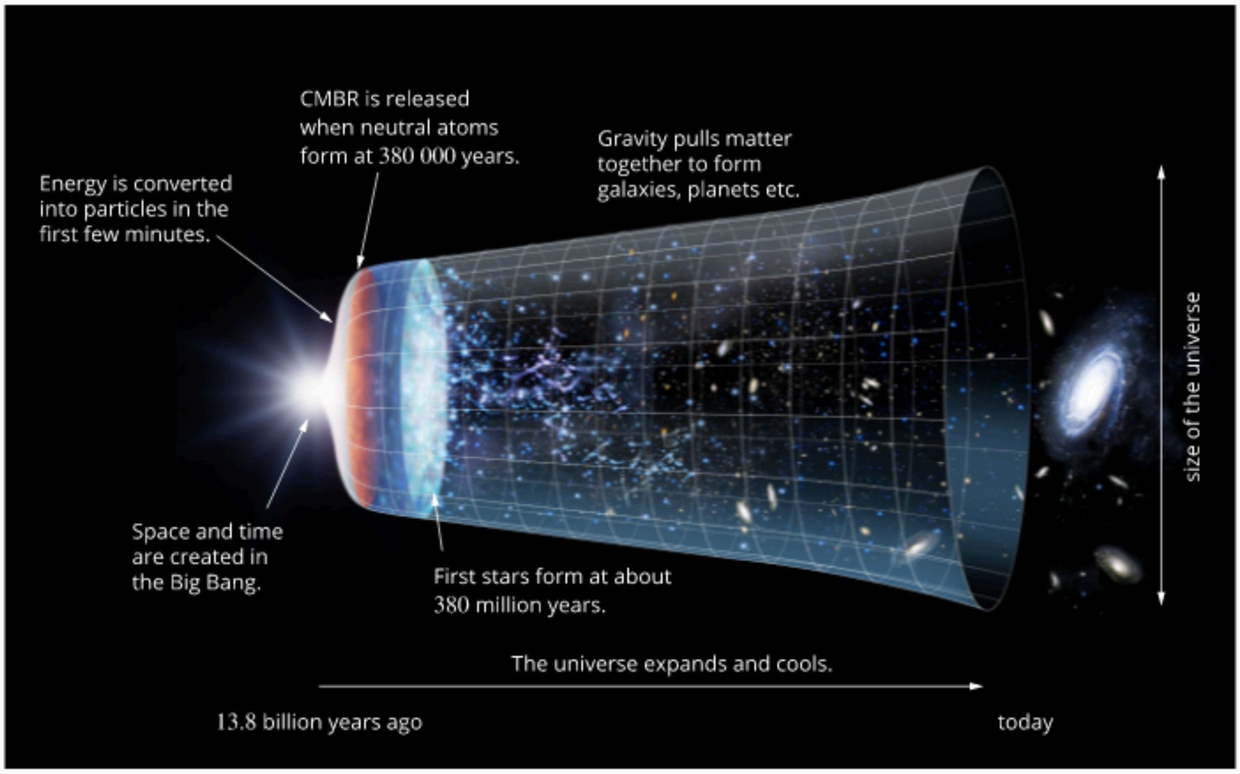
Not according to astronomer Fred Hoyle back in the 1950's. He put forward what became known as the ***Steady State Theory***. His idea was that the Universe really was both infinite and expanding.

If it was infinite, the expansion was not a problem; the 'outer' stars would never reach infinity and so could go on moving away from us forever. This would mean that the Universe was becoming "thinner" - less dense. Hoyle believes that matter is being created all the time at just the right rate to keep the density of the Universe constant.

His theory was disproved when cosmic microwave background radiation was discovered in the early 1960's by Arno Penzias and Robert Wilson.

The ***Big Bang Theory*** was not to be seen as an explosion from a small point in space - it was more correctly an explosion of spacetime. Einstein had already showed that time and space were not separate entities. So the big bang would not have occurred at some point in time any more than it would have occurred at a point in space. ***Time, space and matter were all created together***.

(For more detail, Ch. 9.4 of the text.)



9.4.14 • The 13.8 billion year history of the universe. (Not to scale!)

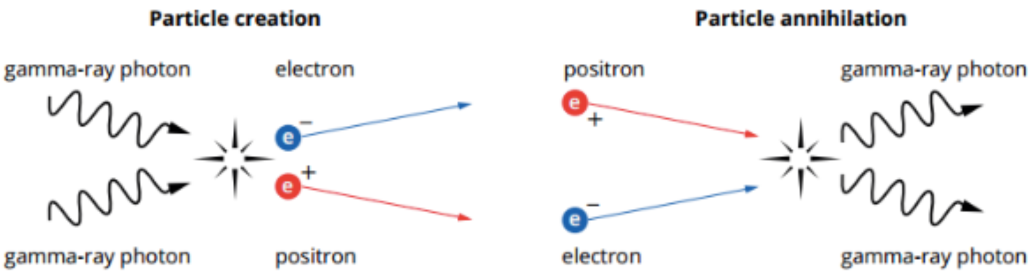


FIGURE 9.4.12 Electron–positron pairs were rapidly going in and out of existence in the early universe. As they met they annihilated, producing more gamma-ray photons, but the photons could also produce the electron–positron pairs.

When was the bang?

If Hubble's law is right, Physicists need to run the rate of expansion backwards to see when all the galaxies were in the same place.

$$v_{galaxy} = \frac{d}{t_{universe}} = H_o d$$

$$\Rightarrow \quad t_{universe} = \frac{d}{H_o d}$$

$$\Rightarrow \quad \boxed{t_{universe} = \frac{1}{H_o}}$$

Useful to know.

Example

What is the age of the Universe if the Hubble constant is taken as 70 km s⁻¹ Mpc⁻¹?

Solution

The age of the Universe is the reciprocal of the constant:

$H_o = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$

$$T_{Universe} = \frac{1}{H_o} = \frac{1}{70}$$
$$= 1.43 \times 10^{-2} \text{ s Mpc km}^{-1}$$

The expression for the constant includes two different units for distance, km and Mpc, so we need to convert Mpc to km. Given that 1 pc = 3.086×10^{16} m:

$\text{pc} = 3.086 \times 10^{13} \text{ km}$

$\text{Mpc} = 3.086 \times 10^{19} \text{ km}$

Convert to km.

$$T_{Universe} = 1.43 \times 10^{-2} \text{ s Mpc km}^{-1}$$
$$= (1.43 \times 10^{-2})(3.086 \times 10^{19})$$
$$= 4.41 \times 10^{17} \text{ s}$$
$$= \frac{4.41 \times 10^{17}}{365.25 \times 24 \times 60 \times 60}$$
$$= 1.40 \times 10^{10} \text{ years}$$

Note the year!

That is, 14 billion years.

PHYSICS IN ACTION

The start of the universe

Throughout recorded history, scientists have always sought to find an answer to how the universe began, as well as if, when and how it will end. Astronomers have constructed hypotheses called cosmological models to try and find the answer. In the 1940s, the **Steady State theory** and the **Big Bang theory** were the two contenders.

Steady State theory

Logic tells us that if space is expanding, then at some time in the past it must have been just a dot. However, in the 1950s, astronomer Fred Hoyle didn't agree and he put forward what became known as the Steady State theory.

He suggested that the universe was:

- infinite—the 'outer' stars would never reach infinity and so could go on moving away from us forever
- expanding—matter is being created all the time at just the right rate to keep the density of the universe constant.

This was not really such an outrageous idea. Quantum mechanics had already suggested that matter was less 'permanent' than previously thought.

Big Bang theory

At the time, the alternative was Hubble's theory that galaxies outside the Milky Way were moving away from us. This therefore implied that at an instant in time, the universe and everything in it was contained in a single point. In other words, everything was created from nothing in one 'big bang'.

At the moment of this big bang, all of the matter and energy in our present universe was packed together in some infinitely small region, which rapidly expanded. Based on understandings from nuclear and particle physics, the Big Bang theory predicts that energy was converted into matter in the early universe. The universe rapidly expanded and cooled. Millions of years later this matter began to condense into galaxies as the universe continued to expand.

It was in fact Hoyle who first used the expression 'big bang' to describe a universe that started off from a tiny dot. At the time, he was intending to be derogatory, but the name stuck.

Steady State theory vs Big Bang theory

The astrophysicists of the day really were caught in a dilemma. Either way they had to accept the unacceptable—that matter just came into being out of nothing. It seemed impossible to resolve this dilemma. If Hoyle was right, then where should scientists look for the new matter being created? If the universe really started with a big bang, how could they possibly know? The observational evidence, an expanding universe, was consistent with both theories.

If matter was being created at a steady rate, it was not too difficult to calculate that about two or three atoms of hydrogen would need to be created every day in a volume about the size of a large sports arena to account for the observed density and expansion. That didn't seem all that impossible, but there was not much point in trying to look for it!

On the other hand, if the Big Bang idea was correct, what difference might it make to what you can see around you now? Not much, but the universe would have been denser at an earlier time. It seemed that only some means of looking back in time could resolve the problem.

FIGURE 9.4.8 In the 1950s, the British astronomer Sir Fred Hoyle proposed the Steady State theory in opposition to the Big Bang theory.

Physics in action — The future of the Universe

If the Universe is expanding, will it go on expanding forever, or will it eventually slow and perhaps start collapsing again? Naturally this question has fascinated astrophysicists ever since the idea of the big bang was suggested. If a rocket is fired upwards, it may reach a maximum height and fall back, or it may have enough energy to escape from the Earth and sail on into space forever. Or it may have *just* enough energy to escape but none left over for space adventures.

The question the astrophysicists asked was 'Did the Universe have enough energy to overcome the mutual gravitational attraction of all its mass and expand forever?' Or would gravity eventually win out and pull it all back into a 'big crunch'—and perhaps start the cycle all over again with another big bang? Actually, because of Einstein's theory of general relativity, which links space, matter and gravity, the question was put a little differently. Mass, Einstein said, distorts space. We see the effects of the 'curved space' as the effects of gravity.

In the language of relativity the question becomes whether space has positive, negative or zero curvature.

Positive curvature implies that there is enough matter in the Universe to fold it in on itself—producing a closed universe that will eventually collapse to the big crunch. Negative curvature is the opposite—space curves outwards in an open-ended way and the expansion can continue forever. In between positive and negative curvature there is the 'flat' universe—equivalent to the expansion *just* being sufficient to go on forever, but only just. These three possibilities are sometimes represented by the two-dimensional analogies shown in Figure 6.80.

There are different approaches to answering this question. One is to try to measure all the mass in the Universe and see if it is sufficient to curve space 'inwards'. The difficulty here is to detect the mass. We have already seen that most of the mass of the Universe appears to be so called 'dark matter', which by its nature is very hard to detect. Another approach

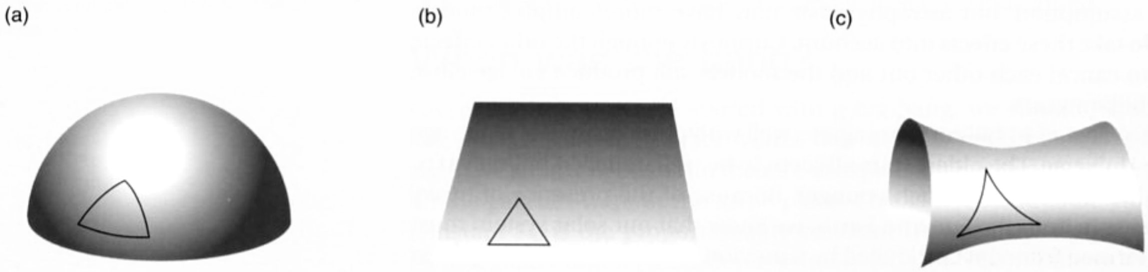


Figure 6.80

These diagrams are two-dimensional analogies for what is really three-dimensional space—or four, if time is included! In (a), positive curvature, the surface eventually folds in on itself, while the negative curvature of (c) can go on expanding forever. Surface (b) is the in-between 'flat' case.

is to measure the rate of expansion at different times in the history of the Universe to see if it is slowing down. This can be done by looking at galaxies that are vast distances from us—the further away, the further back in time we are looking.

It is the cosmic microwave background radiation (CMB) that has given us our best clue to the answer. The CMB has been travelling through space since the time of the hot early universe and has therefore been affected by the space it has travelled through. It is possible to simulate the effects of positive, zero or negative space curvature on the radiation and then compare these with the actual radiation detected by satellites such as WMAP. The differences are found in the very small irregularities we discussed earlier. The results of this research seem fairly clear—the CMB pattern is consistent with a flat universe; that is, zero curvature.

Given that the Universe is flat, the total amount of mass and energy in it can be calculated (the reverse of the process of trying to determine if it is flat from the amount of mass). Remember at this point that Einstein said mass and energy are inextricably linked ($E = mc^2$). It turns out that even with the estimated amount of dark matter, there is still a huge amount of missing 'mass-energy'. Visible matter accounts for about 4% and dark matter for about 26%. This would appear to leave 70% of the total mass-energy content missing. However, you will not be surprised to find that astrophysicists now think they know where it is!

Perhaps the most surprising results have come from the studies of very distant galaxies. They have suggested that rather than slowing down, the rate of expansion of the Universe has been accelerating. At first this seemed to be quite contrary to all expectations! Even our spacecraft with plenty of energy to escape the Earth will still slow down as gravity tugs on them. Similarly, it was thought that the mass of the Universe would eventually slow its expansion rate. However, when this acceleration was put together with the picture of a flat universe, as well as the implications of Einstein's 'cosmological constant', a new idea emerged. It seemed that the missing 70% of the mass-energy content of the Universe might be in the form of what has been called 'dark energy'. This energy, it is thought, is what is actually accelerating the expansion—it is a sort of 'anti-gravity'. As the effect of gravity (space curvature) becomes less with the increasing size of the Universe, the effect of the dark energy becomes more significant and the acceleration increases.

So how will the Universe end? Perhaps unfortunately, it does not seem that it will end with a big crunch after all. At least that may have given rise to another big bang. Rather, as everything accelerates apart it will get thinner and thinner until everything is so far away that it will become a very boring place with a very dim and uninteresting night sky—not much good for astrophysicists at all! Oh well, perhaps by then we'll have found a way to jump to one of those 'parallel universes' we hear about—one with a different, but no less interesting, set of problems for astrophysicists to solve.

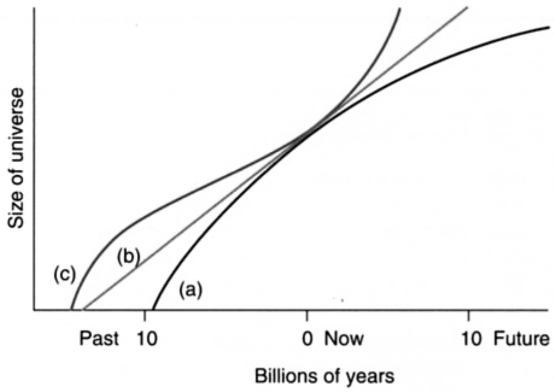


Figure 6.81

(a) This curve is for a flat universe without dark energy, and until recently seemed the most likely model. (b) This curve shows the expansion if we simply assume the Hubble constant has been constant from the beginning. (c) This curve is the latest model, which takes account of dark energy.